Vol 13 Chapter 1 — Biology as Substrate Macromolecular Behavior

Keeper (author pass — deep math/physics revision)

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Chapter 1 — Biology as Substrate Macromolecular Behavior

Level 1 — one sentence

Biology is chemistry at scale — macromolecules (DNA, RNA, proteins, lipids) forming self-replicating systems — emerging from substrate when chemical-scale K-type configurations couple into hereditary information-carrying replicators with population-level Darwinian dynamics.

Level 2 — graduate-physicist precision

1.1 Biology in scale hierarchy

- Atomic (10^{-10} m): Vol 5 K-types
- Molecular (10^{-9} m): Vol 12 bonding
- Macromolecular (10^{-9}-10^{-6} m): proteins, nucleic acids — biology starts here
- Cellular (10^{-6}-10^{-5} m): Vol 13 Ch 8 cell biology
- Organism (10^{-3}-10^{0} m)
- Ecosystem (10^{2}-10^{7} m)

Energy scales: kT ≈ 25 meV at 310 K (body temp); ATP hydrolysis ≈ 0.3 eV; covalent bonds ≈ 4 eV (Vol 12 Ch 2).

1.2 The substrate question

Which BST primaries set natural biological scales?

- N = 4 nucleotides (A, T/U, G, C): N_c + 1 = 4
- 64 codons (3-nucleotide): $4^{N_c} = 64$
- 20 amino acids (canonical): $N_c \cdot n_C + g = 9 + 7 = 16\ldots$ or $C_2 \cdot 3 + 2 = 20$ via Vol 6 Casimir-ladder
- 2 strands of DNA helix: rank = 2
- 20 amino acids + 3 stop codons = 23: cf. M_24 Bridge Object family (Vol 11 Ch 6)

Chapters 2-6 develop these substrate ↔ biology correspondences.

1.3 Self-replication

Life requires: - Heritable information (DNA/RNA sequence) - Catalysis (enzymes lower activation energy) - Compartmentalization (lipid membranes) - Energy harvesting (metabolism)

Each is a substrate K-type configuration at biological scale (chemical bond rearrangements supporting replication).

1.4 Darwinian dynamics

Variation + heredity + selection → evolution. Substrate provides the variation (mutation as substrate K-type alteration), heredity (DNA replication), and the environment provides selection.

1.5 K-audit anchors

- Vol 12 Ch 1-6: chemistry from substrate
- Vol 9 Ch 1-12: condensed matter scale (bridges to bio)

Level 3 — 5th-grader accessibility

Biology is chemistry at scale. Bigger molecules (proteins, DNA), self-replicating, evolving. Key numbers: 4 nucleotides (N_c + 1), 64 codons (4^{N_c}), 20 amino acids, 2 DNA strands (rank). Life requires: heritable info + catalysis + compartments + energy. In BST: biology = substrate's behavior in the macromolecular regime where K-types couple into hereditary replicators.

What comes next

Chapter 2 develops the genetic code and BST primaries.

Where to look this up

- Alberts et al., *Molecular Biology of the Cell*
- BST: Vol 12 (chemistry), Vol 9 (condensed matter)